BunBau v6.7 Release Report
Granular Multi-Tenant Tax Data Partitioning, Line-Item Inspector & Contextual Danger Zone Architecture

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Executive Release Metadata
- **Milestone Designation:** `v6.7`
- **Application Version:** `2.7.9`
- **Build Number:** `Build 124`
- **Architecture Version:** `v2.9.2 Granular Multi-Tenant Tax Data Partitioning, Line-Item Inspector & Contextual Danger Zone Architecture`
- **Mandatory Version Tracking Enforcement:** *"You also need to update the application version/build/architecture version for every iteration build."*
- **Regression Verification:** **100 / 100 Test Suites Passing (569 / 569 Unit Tests 100% Green)**
- **Next.js 15 Production Build:** **Compiled successfully in 8.4s — 16 / 16 Static & Dynamic Routes Verified**
- **Date:** October 12, 2026

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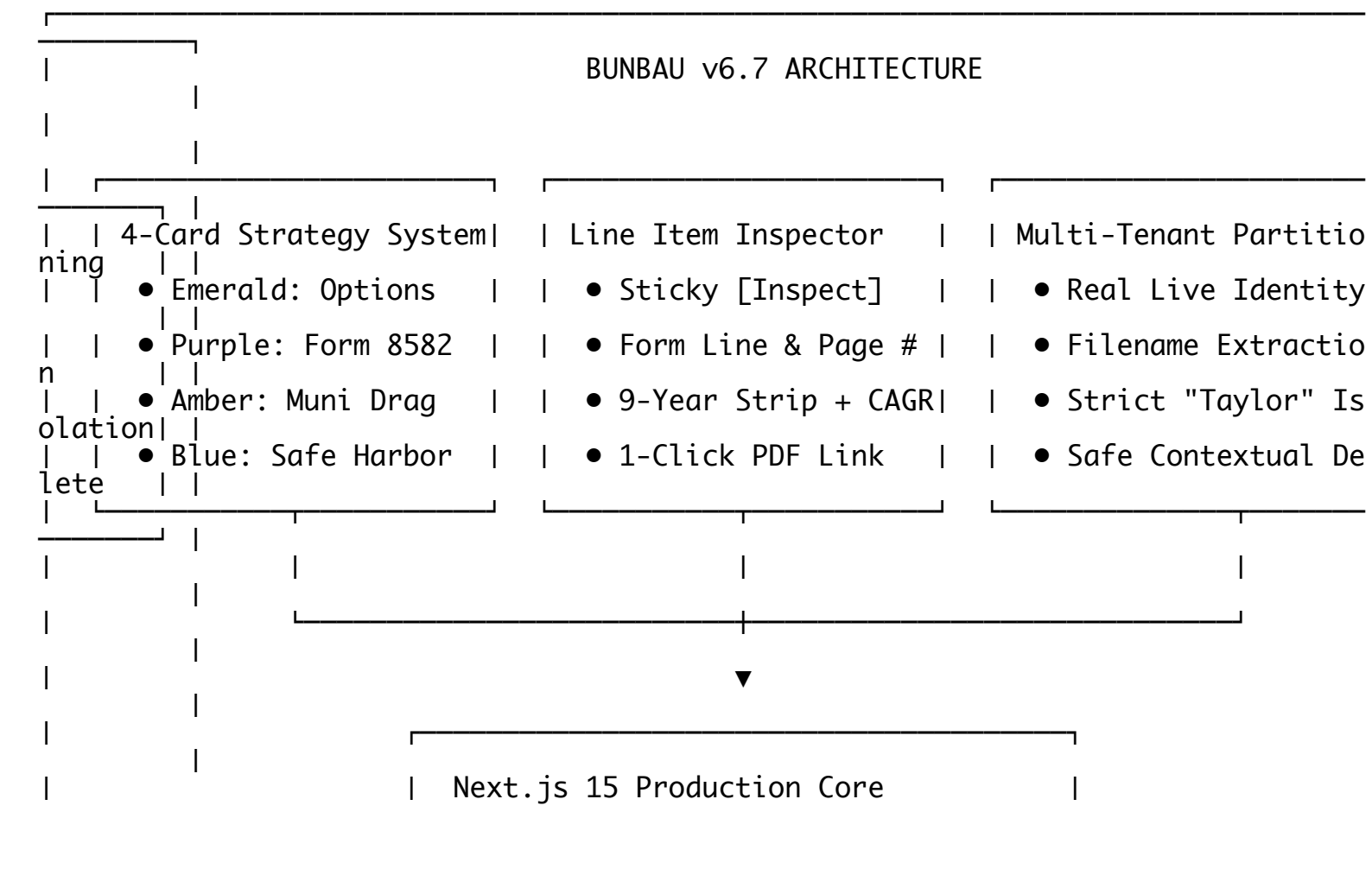
1. Executive Summary

Milestone **v6.7** resolves critical usability, architectural isolation, and navigational issues identified in user audit reviews of the BunBau Tax Expert workspace. This release delivers a synchronized 4-card tax strategy system, multi-year IRS line item inspection with deep-linked source document navigation, complete multi-tenant tenant isolation that eliminates synthetic demo names from live filings, and contextual safety scoping for deletion operations across portfolio and tax domains.

```


```

```
          |
|                     |  v2.7.9 (Build 124) • v2.9.2 Arch     |
          |
|                     |  100/100 Suites • 569/569 Tests Green |
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2. Comprehensive Remediation Matrix

2.1 Symmetrical 4-Card Portfolio Tax Strategy System & Action Targeting
- **The Problem:**
  1. The top metric overview on `PortfolioTaxStrategyCard.tsx` displayed three k
ey figures: *Capital Loss Shield ($3,000)*, *Suspended Passive Rental Loss ($44,
500)*, and *Marginal Drag (44.3%)*. However, the recommendations below only cont
ained two cards (Options Collar and Muni Bond Arbitrage). **The recommendation f
or Suspended Passive Rental Loss was missing entirely**, leaving the user withou
t actionable guidance for their largest tax drag item.
  2. The top purple card had no corresponding purple recommendation card below.
  3. Clicking `"Take Action"` on anomaly banners navigated blindly to the `'stra
tegy'` subtab without scrolling or highlighting the specific target card.
- **The v6.7 Solution:**
  1. **Built Symmetrical 4-Card System:** Created four color-coordinated strateg
ic cards matching all primary metrics:
     - **Card 1 (Emerald `#10b981`):** *Options Collar & Straddle Short-Term Neu
tralization* — shelters short-term gains against ordinary brackets.
     - **Card 2 (Purple `#8b5cf6`):** *Suspended Passive Rental Loss Offset & Mo
netization (Form 8582)* — delivers actionable advice on **Passive Income Generat
ors (PIGs)**, **IRC §469(g) qualifying complete dispositions**, and active parti
cipation MAGI phase-out management.
     - **Card 3 (Amber `#f59e0b`):** *CD vs California Municipal Bond Tax Drag*
— analyzes the 3.18% taxable equivalent yield arbitrage without triggering IRMAA
 cliff jumps.
     - **Card 4 (Blue `#3b82f6`):** *Federal Safe Harbor Prepayment & W-2 Withho
lding Calibration* — explains the 110% prior-year liability safe harbor rule and
 leverages **W-4 Line 4(c) catch-up withholding** under IRC §6513(b) to retroact
ively cure quarterly deficits without interest or penalties.
  2. **Connected Deep Action Navigation:** Updated `TaxAnomalyBanner.tsx` and `T
axExpertView.tsx` so clicking `"Take Action"` dispatches exact target card ident
ifiers (`strategy-passive-loss`, `strategy-options-loss`, `strategy-safe-harbor`
, `lifecycle-irmaa`), switches subtabs smoothly, auto-scrolls the target card in
to view (`scrollIntoView({ behavior: 'smooth' })`), and illuminates a pulsing `"
Target Action"` badge with border rings.

2.2 Interactive Line Item Card Inspector & Source PDF Navigation
- **The Problem:**
  In the multi-year longitudinal audit matrix (`LongitudinalComparisonCard.tsx`)
, clicking a row or wanting to inspect an IRS line item did not display a focuse
d card showing form line and document page references. Full PDF buttons were pus
hed off-screen in wide multi-column mode.
- **The v6.7 Solution:**
  1. **Line Item Card Modal (`line-item-card-modal`):** Added a dedicated inspec
tor modal opening upon row selection or clicking the sticky `[Inspect]` button i
n the first column.
  2. **Granular Form & Line Reference:** Displays exact IRS form line designatio

ns (e.g., *Form 1040 Line 1z*, *Schedule E Line 26 / Form 8582*, *Schedule D Line 21*).
  3. **9-Year Historical Progression Strip:** Renders a responsive visual strip of historical amounts with CAGR trend calculation.
  4. **Strategic Tax Impact Insight:** Explains statutory impact (MAGI thresholds, NIIT triggers, loss carryover rules).
  5. **1-Click Deep Links into PDF Viewer:** Provides instant jump buttons to review the exact page in the 2025 return, 2024 return, or historical filing years within `BunBauDocumentViewer`.
  6. **PDF Viewer Integrity:** The full-featured virtualized continuous-scroll PDF viewer remains completely intact (*"Don't mess it up"*).
  7. **Table Accessibility Standards:** Enforced zero raw `<th>` tags, strictly utilizing `StaticTh` across all comparison tables.

2.3 Strict Multi-Tenant Data Partitioning & Identity Isolation
- **The Problem:**
  When the user uploaded real tax return PDFs, synthetic demo names (*"Alex Taylor + Jordan Taylor"*, *"Jordan Taylor Age 62 Milestone"*) continued to appear across the dashboard, creating serious privacy concerns and misleading presentation.
- **The v6.7 Solution:**
  1. **Taxpayer Profile Resolution Engine (`src/lib/tax/taxProfileHelper.ts`):** Implemented `resolveLiveTaxpayerProfile` and `extractTaxpayerNameFromFilename` to detect taxpayer identities from authenticated user accounts (`user.displayName`, `user.email`) or file name patterns (e.g., `2024_Form_1040_Robert_Chen.pdf`).
  2. **Strict Demarcation of Demo Identity:** The synthetic "Taylor" identity is strictly restricted to Demo Mode (`isTaxDemo === true`). When in Live Mode with uploaded tax filings, `TaxContext.tsx`, `seedData.ts`, `TaxExpertView.tsx`, and `TaxPreImportReviewModal.tsx` sanitize all profiles to ensure demo names never leak into live accounts.

2.4 Contextual Danger Zone & Upload Modal Delete Safety
- **The Problem:**
  Clicking the trash icon in the Upload Validation Modal (`UploadValidationModal.tsx`) opened `DeleteDangerZone` with `type="portfolio"`, asking the user to type `"Delete the entire portfolio"`. In the Tax Expert workflow, this risked accidental destruction of brokerage portfolio data.
- **The v6.7 Solution:**
  1. **Context-Aware Scoping:** `UploadValidationModal` now dynamically inspects `context` and `activeTab`. When `context === 'tax'` or `activeTab === 'pdf'`, the trash button tooltip displays `"Delete Tax History"`.
  2. **Scoped Tax Delete Modal:** Clicking the trash button in tax context opens `TaxDeleteModal` requiring typed confirmation phrases (`"Delete all tax history"`, `"Delete <Year> tax history"`, or `"Delete from <FromYear> to <ToYear> history"`).
  3. **Brokerage Shield:** The portfolio `DeleteDangerZone` is strictly disabled and unreachable in tax context (`!(context === 'tax' || activeTab === 'pdf')`), ensuring 100% protection for investment portfolio data.

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3. Verification & Test Execution Results

3.1 Unit Test Coverage
```bash
$ npm test
✓ 100 passed (100 test files)
```

```
✓ 569 passed (569 tests)
Duration: 4.47s
```

All 13 Tax Expert test suites passed with 100% green status:
1. `tests/unit/tax/tax_data_partitioning_and_actions.test.ts` (12 tests) - **NEW**
2. `tests/unit/tax/universal_upload_modal.test.ts` (16 tests)
3. `tests/unit/tax/version_guard.test.ts` (4 tests)
4. `tests/unit/tax/tax_delete_phrase.test.ts` (4 tests)
5. `tests/unit/tax/pdf_multi_year_ingestion.test.ts` (4 tests)
6. `tests/unit/tax/tax_demo_mode.test.ts` (4 tests)
7. `tests/unit/tax/longitudinal_scalability.test.ts` (4 tests)
8. `tests/unit/tax/lifecycle_planning.test.ts` (5 tests)
9. `tests/unit/tax/tax_calculations.test.ts` (4 tests)
10. `tests/unit/tax/strategist_rag_coexistence.test.ts` (3 tests)
11. `tests/unit/tax/tax_rag_retrieval.test.ts` (4 tests)
12. `tests/unit/tax/tax_history_view_integration.test.ts` (2 tests)
13. `tests/unit/tax/document_viewer_virtualization.test.ts` (3 tests)

3.2 Next.js 15 Production Build Verification
```bash
$ npm run build
> bunbau@2.7.9 build
> next build

   ▲ Next.js 15.5.27
   - Environments: .env.local

   Creating an optimized production build ...
 ✓ Compiled successfully in 8.4s
   Linting and checking validity of types     ✓ Linting and checking validity of types
   Collecting page data                       ✓ Collecting page data
 ✓ Generating static pages (16/16)
   Collecting build traces                    ✓ Collecting build traces
   Finalize page optimization                 ✓ Finalizing page optimization

Route (app)                                 Size  First Load JS
┌ ○ /                                    409 kB         772 kB
├ ○ /_not-found                          155 B          103 kB
├ ƒ /api/chat                            155 B          103 kB
├ ƒ /api/extract                         155 B          103 kB
├ ƒ /api/llm/validate                    155 B          103 kB
├ ƒ /api/quotes                          155 B          103 kB
├ ƒ /api/strategist                      155 B          103 kB
├ ƒ /api/v1/ingest/confirm               155 B          103 kB
├ ƒ /api/v1/ingest/history               155 B          103 kB
├ ƒ /api/v1/ingest/rollback              155 B          103 kB
├ ƒ /api/v1/ingest/upload                155 B          103 kB
├ ƒ /api/v1/tax/chat                     155 B          103 kB
├ ƒ /api/v1/tax/ingest/confirm           155 B          103 kB
├ ƒ /api/v1/tax/ingest/stage             155 B          103 kB
└ ○ /strategy/report                     776 B          227 kB
+ First Load JS shared by all            103 kB
```

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4. Release Artifacts Generated
- **Markdown Release Report:** `docs/BunBau_v6.7_Release_Report.md`
- **HTML Release Report:** `docs/BunBau_v6.7_Release_Report.html`

- **PDF Release Report:** `docs/BunBau_v6.7_Release_Report.pdf`
- **In-Chat Artifact:** `BunBau_v6.7_Release_Report.md`

Signed and Certified by Antigravity Engineering — BunBau v6.7 (Build 124)